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Task Description:	Hematopoietic stem cells (HSC) comprise less than 0.1% of the bone marrow (BM) of adults, yet they produce all blood cells that are responsible for constant maintenance and immune protection of the body. In addition to the requirement for high cell output, the concentration of the different blood cell lineages produced are precisely controlled to allow not only preservation of normal homeostasis, but also the prompt response to specific stimuli and demands. Previous studies have suggested that exposure to galactic cosmic radiation (GCR) and solar particle event (SPE) radiation that occur or may potentially occur during space flight may increase the likelihood of carcinogenesis, with leukemias being one of the most frequent radiogenic cancers, and the ones with the smallest latency period. We hypothesize that GCR and SPE radiation can have a deleterious impact not only at the level of the mature immune cells and their committed progenitors, as is currently thought, but at the earlier developmental stage of the HSC. Genetic lesions induced by this exposure can lead directly to leukemic transformation, and may also cause these cells to generate a functionally compromised immune system lacking the ability to effectively fight challenges such as pre-leukemic transformations and viral/bacterial infections. As such, we hypothesize that astronauts would be doubly at risk, due to both the potential for induction of leukemia by SPE/GCR radiation, and the reduced ability of the astronauts' immune systems to identify and eliminate malignant cells or pathogens in the body. In the present proposal, we will use a variety of both in vitro cell culture and in vivo transplantation systems to rigorously test the novel hypothesis that combined SPE/GCR radiation can directly induce DNA damage and mutations within primitive HSC, and can also contribute to defective hematopoiesis/immunity through so-called tissue bystander effects, by likewise damaging the BM microenvironmental/niche cells (MSC) that regulate hematopoiesis. The overall goal of this proposal is, therefore, to utilize unique model systems developed in our laboratory to precisely and reproducibly define the effects of SPE/GCR radiation on the human hematopoietic and immune systems, at both the molecular and functional levels, thereby defining the possible risk of leukemogenesis as a result of exposure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed studies directly address NASA's Radiation Program General Focus by studying cellular differentiation, genetic sensitivity or resistance, and immune function, all of which NASA has identified as being of particular importance. Moreover, the proposed approach of both creating a human BM microenvironment in immunodeficient mice and then repopulating that niche with human MSC and HSC that have been exposed to both SPE-like protons and low fluence heavy ion components of GCR directly addresses another of NASA's stated priorities which is "to develop new 3D human cell culture or transgenic mouse models of human tissues improving our ability to extrapolate estimates of cancer." With respect to the specific goals of this Announcement, our studies to employ human HSC and human BM microenvironmental cells directly address one of the stated high priority areas: "research that considers adult stem cells and the stem cell niche." In addition, our studies to investigate SPE/GCR-induced chromosomal aberrations (CAs), alterations in cancer-related SNPs and miRNAs, and mutations within coding sequences, assessed through whole exome sequencing (WES), directly respond to another stated high priority goal of elucidating "how genetic and epigenetic factors modulate biological responses to space radiation." Our intentional use of HSC and MSC samples from donors of both genders is directly responsive to NASA's statement "research on gender effects … are of interest." Thus, in combination, our proposed studies investigate multiple different cellular/molecular avenues of radiation carcinogenesis, and directly address another stated high priority for "research that will improve the understanding of radiation quality effects using two or more cancer development processes that are important in radiation carcinogenesis." With respect to NASA's identified "Radiation Carcinogenesis-Specific Gaps," our proposed studies address Cancer 1, as they include development of an experimental model of tumor development for leukemia, which, with this "humanized" model, will allow more direct extrapolation of human risk projection. The proposed genomics studies (CAs, miRNA, WES, and SNP arrays) and the inclusion of human donors of both genders (with an age range of typical astronauts) also directly addresses Cancer 5, which requests application of "models of cancer risk … to reduce the uncertainties in individual radiation sensitivity including genetic and epigenetic factors from GCR and SPE," and Cancer 6, which requests application of "models of cancer risk … to reduce the uncertainties in age and gender dependence of cancer risks from SPEs and GCR."
	Progress NASA-Funded Project: In ground-based studies performed at NASA Space Radiation Laboratory (NSRL) and Wake Forest, we have spent the last 4 years assessing the risk of leukemogenesis from space radiation by examining the effects of simulated SPE/GCR exposures on human hematopoiesis, testing the hypothesis that the quality and quantity of cells being produced can be altered not only from direct exposure of HSC themselves, but also via indirect, biological bystander effects arising from exposure of MSC within the critical hematopoietic niche. Human HSC and MSC were immunomagnetically-sorted with antibodies to CD34 and Stro-1 respectively and exposed to: A) sham irradiation; B) 1 Gy of 137Cs gamma-irradiation (LET = 0.91 keV/μm); C) 1 Gy of 50 MeV protons (primary SPE energy; LET = 1.26 keV/μm); D) 20 cGy of 1 GeV/n 56Fe ions (model GCR radiation; LET = 151 keV/μm); or E) 1 Gy of 50 MeV protons followed 15 min later by 20 cGy of 1 GeV/n 56Fe ions at NSRL. Immediately following irradiation, aliquots of HSC and MSC were cultured for chromosomal aberration (CA) analyses and DNA/RNA isolations, one aliquot of HSC was immediately assessed for functionality/ differentiative potential using the highly sensitive and quantitative, luminescence-based Hemogenix HALO® assay, and the remainder of the cells were placed in an MSC-based HSC maintenance/expansion culture system we previously reported, testing 4 different combinations: 1) unirradiated HSC plated over unirradiated MSC; 2) unirradiated HSC plated over irradiated MSC; 3) irradiated HSC plated over unirradiated MSC; and 4) irradiated HSC plated over irradiated MSC. Hematopoietic cells were then collected after 7 days, and methylcellulose or HALO® assays were used to assess the impact of irradiation of the HSC, the MSC, or both cell types on hematopoietic colony-forming potential. Direct Effects on Human HSC: As we recently reported (see Bibliography for Leukemia publication), our methylcellulose and HALO® assay data showed that gamma-rays and 50 MeV protons exerted fairly similar effects on HSC function, both reducing by 50–80% the ability of HSC to form both more primitive (HPP-CFC and CFU-GEMM) colonies, as well as more differentiated myeloid (CFU-GM), erythroid (BFU-E), megakaryo-cyte/platelet (CFC-Mk), and lymphoid (CFC-T and CFC-B) colony types. The marked deleterious effect of protons on the ability of human HSC to give rise to cells of the T and B lymphoid lineages is of particular note, given the body of published work showing that astronauts, even on fairly short missions, exhibit pronounced immune alterations. In contrast to the result seen with gamma-rays and 50 MeV protons, 1 GeV/n 56Fe ions had much less pronounced effect on HSC colony-forming potential, exerting no significant effect on their ability to give rise to the most primitive HPP-CFC colonies, but reducing

the colony-forming potential in the remaining lineages by 20–40%, depending upon the specific colony type. Exposing sequentially to protons then iron ions, however, proved to be highly deleterious to HSC function, reducing the ability of HSC to produce all colony types by 80–90%. This decrease was, in all cases statistically different than the effects seen with either iron ions alone ($p<0.01$) or protons alone ($p<0.05$).

Indirect/Biological Bystander Effects on Hematopoiesis following Exposure of the Niche to SPE/GCR: We next used HALO® assays to analyze HSC functionality following co-culture with normal or irradiated MSC to assess whether a microenvironment that had been damaged by exposure to protons and/or iron ions would negatively impact their hematopoietic potential. Irradiation of MSC with gamma-rays or protons produced similar effects on the most primitive HSC, reducing HPP-CFC output 30–40%. Importantly, irradiation of both HSC and MSC produced a negative additive effect on the most primitive HSC, reducing HPP-CFC output ~75%. The results with 56Fe ions were far more unexpected, as irradiation of MSC alone reduced HPP-CFC output 30%, yet co-culture of both irradiated HSC and MSC somehow restored the HPP-CFC potential.

Additive Negative Effects on Hematopoiesis Following Sequential Exposure to SPE then GCR Radiation: Perhaps the most remarkable results came from the sequential treatment with protons and iron ions. While sequential treatment reduced HPP-CFC output 70%, treating the MSC with this sequential dosing regimen actually increased HPP-CFC output of the co-culture roughly 20%. Unexpectedly, however, co-culturing HSC and MSC that had both been sequentially exposed to both ions resulted in an 80% reduction in HPP-CFC output. Very similar results were obtained when we examined the next most primitive colony type, CFU-GEMM, and more differentiated myeloid CFU-GM. Striking results were also obtained when we examined the effect proton and iron ion irradiation of the MSC had upon co-cultured HSC's ability to generate megakaryocyte and lymphoid (CFC-T and CFC-B) colonies.

Alterations to Human Hematopoiesis Assessed in vivo: We conditioned NOD/SCID/IL-2rgamma null (NSG) mice by administering 2 sequential doses of busulfan spaced 24h apart, as we previously described. Conditioned NSG mice were then transplanted with 2x10(5) human HSC that had been exposed to one of the following irradiation schemes (n=9 per group) at the NSRL: 1) sham irradiation; 2) 1 Gy of 137Cs gamma-irradiation; 3) 1 Gy of 50 MeV protons; 4) 20 cGy of 1 GeV/n 56Fe ions; or 5) 1Gy of 50 MeV protons followed immediately by 20 cGy of 1 GeV/n 56Fe ions. Mice were then sacrificed at 4–6 months post-transplant, their spleen and bone marrow harvested, and analyzed by multicolor flow cytometry to ascertain the levels of human hematopoietic engraftment and assess whether exposure to SPE/GCR altered the engraftment potential and/or differentiative capacity of the HSC. In addition to affecting overall levels of engraftment, treatment with protons, 56Fe ions, or sequential exposures also led to pronounced skewing of the differentiative potential of the human HSC, as shown by the relative lineage distribution of the engrafted human cells in the bone marrow and spleens of the transplanted mice. As we reported in Leukemia (2017), commitment to the erythroid lineage was significantly impaired following exposure of human HSC to all radiation types tested, and there was a significant expansion of both the T and B lineages. In addition to this lineage skewing, 5 of the 14 recipients of 56Fe ion-irradiated human HSC (obtained from 2 different healthy donors) exhibited dramatic splenomegaly, with one mouse's spleen reaching ~30X the normal size. To ascertain the cause of splenomegaly, spleens were analyzed by histopathology and immunophenotyping. The spleen exhibited an abnormal morphology, populated largely by cells with a rounded, highly monotonous appearance consistent with lymphoid cells. Immuno-histochemistry with human-specific anti-bodies confirmed that the vast majority of the cells populating the spleen were human, and they expressed CD3 and CD7. Many of these cells also expressed terminal deoxynucleotidyl transferase (TdT), a marker frequently associated clinically with human lymphocytic leukemias. The absence of any staining in an untransplanted NSG mouse spleen stained with the same panel of antibodies verified their specificity. The observed histopathology and immunophenotyping led a team of clinical hematopathologists to concur that human T-ALL developed in these mice. To our knowledge, this represents the first report of hematological malignancy as a result of exposing human HSC to mission-relevant doses of HZE ions. Molecular basis for biological bystander effects: To begin understanding how exposure of the MSC to SPE/GCR was altering their ability to support normal hematopoiesis, we collected RNA from the MSC, prior to and immediately following exposure to gamma, SPE, or GCR. We also collected RNA after the MSC had been passed 5 times in culture, to determine what long-term/stable alterations in gene expression took place as a result of exposure to these various radiation species. Analyses by heat map and principal component analysis (PCA) have revealed very tight clustering of cells treated with each radiation type, showing significant differences exist between not only irradiated cells and their respective pre-irradiation control, but also between the same cells exposed to gamma-rays, SPE, or GCR.

Task Progress:

Setting our analysis to only consider genes whose differential expression is significant at $p<0.05$, we have observed differential expression in as many as 287 genes, depending which conditions are being compared. Analysis of this transcriptome data is still ongoing. Cytokine arrays performed on the MSC-conditioned media have already confirmed many of these results at the protein level, and have shown that multiple pathways involved in immune response and inflammation are altered as a result of exposure to SPE/GCR radiation.

Our pathway analysis results to-date show that at 24h post-irradiation, the gene expression profile of MSC is markedly altered by exposure to each radiation type, with proton and iron ion irradiation perturbing significantly more cellular pathways than gamma-irradiation. Moreover, it is particularly interesting to note that, while gene expression changes following exposure of human MSC to 1 Gy of 137Cs gamma-irradiation largely return to normal pre-irradiation levels following 5 passages in culture, the changes induced by exposure to either 1 Gy of 50 MeV protons (SPE) or 20 cGy of 1 GeV/n 56Fe ions (GCR) persist in the MSC, even after 5 culture passages. These results thus suggest that the epigenetic changes induced by SPE/GCR radiation are perpetuated following cell division, and are passed on to the resultant daughter cells, potentially resulted in long-term alterations to the BM micro-environment and its ability to support normal hematopoiesis.

Testing of a counter-measure and development of a nanoparticle-based vehicle to enable oral administration: Importantly, our ongoing functional and "'omic" analyses of proton/HZE-treated MSC and HSC have provided a wealth of data on molecular signaling pathways that are likely to be involved in some of the observed space radiation effects on hematopoiesis, providing us possible targets for appropriate counter-measure development. Based on these data, we have begun testing a well-studied dietary additive (curcumin) for its ability to protect the human hematopoietic system from the deleterious effects of SPE/GCR radiation.

Curcumin is a natural bioactive compound derived from the turmeric Curcuma longa, which exerts anti-proliferative and pro-apoptotic effects in a wide variety of cellular and animal models of cancer. In fact, curcumin is being used as a radio and chemo-sensitizer in multiple types of cancer, including colon, prostate, and leukemia. The potential of using curcumin as an antioxidant to reduce the cellular damage induced by ionizing radiation has been studied in limited model

systems. Our studies are designed to fully characterize the window of protection against low-dose, low-dose-rate irradiation, including therapeutic potential when administered after irradiation. Curcumin has been reported to protect various study systems, in vitro and in vivo, against the deleterious effects induced by ionizing radiation, and to enhance the therapeutic effect of radiation when used to treat a variety of tumors. In preliminary studies conducted during NSRL 15A and 15C, that it appears to have the ability to serve as an easily administered countermeasure to mitigate the deleterious effects of SPE/GCR on the human hematopoietic system, significantly restoring both primitive (HPP-CFC) and myeloid-committed (CFU-GM) colony-forming potential to primitive human HSC that have been exposed to a high LET 56Fe ions, and partially restoring sSPE proton-induced suppression of myeloid colony (CFU-GM)-forming potential.

Manuscripts during funding period:

1. Rodman C, Almeida-Porada G, George SK, Moon J, Soker S, Pardee T, Beaty M, Guida P, Sajuthi SP, Langefeld CD, Walker SJ, Wilson PF, Porada CD. In vitro and in vivo assessment of direct effects of simulated solar and galactic cosmic radiation on human hematopoietic stem/progenitor cells. Leukemia 2017 Jun;31(6):1398-1407. doi: 10.1038/leu.2016.344. Epub 2016 Nov 24.

2. Almeida-Porada G, Rodman C, Moon J, George S, Soker S, Pardee T, Guida P, Sajuthi SP, Langefeld CD, Walker SJ, Wilson PF, Porada CD. Biological bystander effects on human hematopoiesis due to exposure of bone marrow niche to simulated solar and galactic cosmic radiation. Manuscript in final preparation for submission to Stem Cell Reports, 2017

Abstracts and Presentations at Meetings during funding period:

1. Almeida-Porada G, Rodman C, George S, John Moon J, Soker S, Pardee T, Beaty P, Guida P, Walker SJ, Wilson PF, Porada CD. Space Radiation-Induced Risks of Leukemogenesis. Abstract accepted for a Poster Presentation at the 6th Annual International Space Station Research and Development Conference, Washington, DC, 2017.

2. Rodman C, Almeida-Porada G (co-1st author), George S, John Moon J, Soker S, Pardee T, Beaty P, Guida P, Walker SJ, Wilson PF, Porada CD. Space Radiation-Induced Risks of Leukemogenesis. Selected for oral presentation at NASA Human Research Program Investigator's Workshop, Galveston, TX, 2017.

3. Rodman C, Almeida-Porada G (co-1st author), George S, John Moon J, Soker S, Pardee T, Beaty P, Guida P, Walker SJ, Wilson PF, Porada CD. Space Radiation-Induced Risks of Leukemogenesis. Selected for oral presentation at American Society for Gravitational and Space Research, Annual Meeting, Cleveland, OH, 2016.

4. Rodman C., Almeida-Porada G., George S., Moon J., Walker S.J., Wilson P., Porada C.D. Role of the Bone Marrow Microenvironment in Space Radiation-Induced Leukemogenesis. Selected for oral presentation at NASA Human Research Program Investigator's Workshop, Galveston, TX, 2016.

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6. Rodman C, Almeida-Porada G, George S, Moon J, Walker SJ, Wilson P, Porada CD. Role of the Bone Marrow Microenvironment in Space Radiation-Induced Leukemogenesis. Radiation Research Society, Annual Meeting, Weston, FL, 2015.

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